MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025) Third SEMESTER Discipline Specific Core Course (DSC) -

MG3DSCCMA200 - AI Techniques for Data Analysis

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks:50**

Part A

Very Short Answer Questions

Answer all questions

1.Differentiate between nominal and ordinal data.
Nominal data: categories without order (e.g., gender). Ordinal data: ordered categories (e.g., satisfaction levels).
Marking: Nominal–1, Ordinal–1

2. List two applications of AI in manufacturing.
Any two: predictive maintenance, quality inspection, industrial robots, demand forecasting. 1 mark each.

3. Write a Python example code using the break statement in a loop.
Example:

```
for i in range(5):
   if i==3:
      break
   print(i)
```

Marking: correct loop–1, break usage–1

4. Give two differences between Numpy array and Python list.

NumPy arrays are homogeneous and faster; lists can store mixed types and are slower.

5.Define Pandas Index. What is the default index in a Pandas DataFrame?
Pandas Index: labels identifying rows. Default index: RangeIndex starting from 0.

Part B

Short Answer Questions

Answer any 5 questions

Each question carries 5 marks

6 Write briefly about some popular data analytics tools with their key uses.

Data analytics tools are software applications used to collect, process, analyze, visualize, and interpret data for decision-making.

Example: Python, R-programming etc.

Introduction/definition–1, key concepts–2, examples–2 Marks

7 .Explain the growth of AI from 2010 to present.

Artificial Intelligence (AI) has grown rapidly due to advances in computing power, big data, and machine learning algorithms.

2010–2015

- Rise of deep learning.
- Improved image and speech recognition.
- Increased use of GPUs for AI training.

2016–2020

- AI defeated human champions in games (e.g., AlphaGo).
- Rapid growth of self-driving vehicle research.
- AI widely adopted in healthcare, finance, and e-commerce.
- Development of advanced Natural Language Processing (NLP).

2021–Present

- Emergence of Generative AI models.
- Large Language Models (LLMs) capable of human-like text generation.
- AI integrated into education, healthcare, manufacturing, banking, and software development.
- Focus on Explainable AI (XAI), ethical AI, and responsible AI practices.

Introduction/definition–1, key concepts–2, examples/explanation–2.

8.Explain the importance of data structures in Python with examples
Data structures are methods of organizing and storing data efficiently for easy access and manipulation.

Examples

- List: marks = [75, 80, 90]
- Tuple: point = (10, 20)
- Dictionary: student = {"Name":"Anu", "Age":20}
- Set: colors = {"Red","Blue","Green"}

Introduction/definition–2, examples/explanation–4 Marks

9. Discuss relational operators in Python with examples
Relational operators compare two values and return either True or False.

==, !=,<,> etc…

Example:

a = 10

b = 20

print(a == b) # False

print(a < b) # True

print(a != b) # True

Introduction/definition–1, key concepts–2, examples/explanation–2.

10. Describe different methods to detect outliers in a dataset with examples.
An outlier is a data value that differs significantly from other observations.

Methods

1. Box Plot (IQR Method)
2. Z-Score Method
3. Scatter Plot
4. Histogram

Example

Dataset: 10, 12, 14, 15, 16, 100

Here, 100 is an outlier.

definition–1, Methods–2, examples–2. Marks

11. Differentiate the use of line plot and bar plot.

Definition of Line Plot 1-Marks,Definition of bar plot 1-Marks, Any three difference 1 marks each

12. Differentiate between descriptive statistics functions mean(), median(), and mode() in Pandas.

Mean()-Returns the arithmetic average

Median()-Returns the middle value after sorting

Mode()-Returns the most frequently occurring value

definition–3, Any two difference –2 Marks

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13.Explain selective statements in Python with examples.

Selective statements in Python: if, if-else, nested if, if-elif-else. Include syntax and examples.
Concepts–5, syntax–4, examples–6.

14.What is data visualization? Explain the importance of visualization in analytics and discuss different types of charts/graphs with examples.

Data visualization, importance, and charts (line, bar, pie, histogram, scatter, box plot, etc with examples.
Definition–2, importance–4, chart types–6, example-3 marks